RAJALAKSHMI ENGINEERING COLLEGE

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S2

Started on	Friday, 28 August 2026, 2:14 PM
State	Finished
Completed on	Friday, 28 August 2026, 2:15 PM
Time taken	1 min 15 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int bs(int a[], int l, int r, int x) {
    if (l > r) return -1;
    int m = (l + r) / 2;
    if (a[m] == x) return m;
    if (a[m] > x) return bs(a, l, m - 1, x);
    return bs(a, m + 1, r, x);
}

int find(int a[], int l, int r, int n, int x, int *p, int *q) {
    if (l > r) return 0;

    int m = (l + r) / 2;
    int k = bs(a, 0, n - 1, x - a[m]);

    if (k != -1 && k != m) {
        *p = a[m];
        *q = a[k];
        return 1;
    }

    return find(a, l, m - 1, n, x, p, q) ||
           find(a, m + 1, r, n, x, p, q);
}

int main() {
    int n, x, p, q;
    scanf("%d", &n);
    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    if (find(a, 0, n - 1, n, x, &p, &q)) {
        printf("%d\n%d\n", p, q);
    } else {
        printf("No\n");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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